1

Introduction

Chapter Outline

1.1 What Is the Value of Statistics?
Cigarette Smoking Causes Cancer—Tobacco Industry Denies Charges
North Carolina Congressional Districts Gerrymandered—African Americans Slighted
Global Warming—Myth According to the President
1.2 Brief Introduction to the History of Statistics
1.3 General Statistical Definitions
1.3.1 Statistical Notation
1.4 Types of Variables
1.5 Scales of Measurement
1.5.1 Nominal Measurement Scale
1.5.2 Ordinal Measurement Scale
1.5.3 Interval Measurement Scale
1.5.4 Ratio Measurement Scale
1.5.5 Summary of Terms
1.6 Additional Resources

Key Concepts

1. ***General statistical concepts***
   - Population
   - Parameter
   - Sample
   - Statistic
   - Descriptive statistics
   - Inferential statistics
2. ***Variable-related concepts***
   - Variable
   - Constant
   - Categorical variables
   - Dichotomous variables

textbook. We have written the book with the perspective that it is more important to understand concepts than to solve theorems and derive or memorize various and sundry formulas. If you understand the concepts, you can always look up the formulas if need be. If you do not understand the concepts, then knowing the formulas will only allow you to operate in a cookbook mode without really understanding what you are doing.

Third, the calculator and computer are your friends. These devices are tools that allow you to complete the necessary computations and obtain the results of interest. There is no need to compute equations by hand (another reason why we concentrate on the concepts rather than formulas). If you are performing computations by hand, find a calculator that you are comfortable with; it need not have 800 functions, as the four basic operations, sum, and square root functions are sufficient (one of our personal calculators is one of those little credit card calculators, although we often use the calculator on our computers). If you are using a statistical software program, find one that you are comfortable with (most instructors will have you use a program such as **R**, SPSS, or SAS). In this text, we do walk through basic formulas by hand so that you become acquainted with how the statistical program works and the numbers that are used in it. However, we don't anticipate (nor do we encourage) that you make a practice of working statistics by hand. Throughout the text, we use SPSS and **R** to illustrate statistical applications. Although this book is *not* a guide on all things SPSS and **R**, we do try to provide the tools you need to compute the various statistics. We hope that you will supplement what we provide with your own motivation to learn more about software that can assist you in computing statistics.

Finally, this text will take you from raw data to results using realistic examples. The examples may not always be from a discipline that is like the one you are in, but we hope that you are able to transfer or generalize the illustration to an area in which you more comfortable. These examples can then be followed up using the problems at the end of each chapter. Thus, you will not be on your own, but will have the text, a computer/calculator, as well as your course and instructor, to help guide you.

The intent and philosophy of this text is to be conceptual and intuitive in nature. We have written the text so that students who have completed basic mathematical requirements in high school can be comfortable reading the text. Thus, the text does not require a high level of mathematics, but rather emphasizes the important concepts in statistics. Most statistical concepts really are fairly easy to learn because (a) concepts can be simply stated, (b) concepts can be related to real-life examples, (c) many of the same concepts run through much of statistics, and therefore (d) many concepts can be related.

In this introductory chapter, we describe the most basic statistical concepts. We begin with the question, "What is the value of statistics?" We then look at a brief history of statistics by mentioning a few of the more important and interesting statisticians. Then we consider the concepts of population, parameter, sample, statistic, descriptive and inferential statistics, types of variables, and scales of measurement. Our objectives are that by the end of this chapter you will (a) have a better sense of why statistics are necessary, (b) see that statisticians are an interesting group of people, and (c) have an understanding of several basic statistical concepts.

1.1 What Is the Value of Statistics?

Let us start off with a reasonable rhetorical question: "Why do we need statistics?" In other words, what is the value of statistics, either in your research or in your everyday life? As

matters. For instance, a certain type of research, statistical analysis, and set of results are all necessary to make causal inferences about cigarette smoking. Another type of research, statistical analysis, and set of results are all necessary to lead one to confidently state that the districting system is racially biased or not, or that global warming needs to be dealt with. *The bottom line is that the purpose of statistics, and thus of this text, is to provide you with the tools to make important decisions in an appropriate and confident manner using data*. W. Edwards Deming has been credited with bringing quality to manufacturing (e.g., Gabor, 1990), and he once stated, "In God we trust. All others must have data." These are words to live by! After reading this text, you will not have to trust a statement made by some so-called expert on an issue, which may or may not have any empirical basis or validity; you can make your own judgments based on the statistical analyses of data. For you, the value of statistics can include (a) the ability to read and critique articles in both professional journals and in the popular press, and (b) the ability to conduct statistical analyses for your own research (e.g., thesis or dissertation). We hope that this text will guide you in becoming both a better consumer and better producer of statistics. You are gaining skills that you can use to make a contribution to your field and, more important, make the world a better place. The statistical skills you are gaining through this text are powerful. Use them—wisely!

1.2 Brief Introduction to the History of Statistics

As a way of getting to know the topic of statistics, we want to briefly introduce you to a few famous statisticians. The purpose of this section is not to provide a comprehensive history of statistics, as those already exist (e.g., Heyde, Seneta, Crepel, Feinberg, & Gain, 2001; Pearson, 1978; Stigler, 1986). Rather, the purpose of this section is to show that famous statisticians are not only interesting, but are human beings just like you and me.

One of the fathers of probability (see Chapter 5) is acknowledged to be Blaise Pascal from the late 1600s. One of Pascal's contributions was that he worked out the probabilities for each dice roll in the game of craps, enabling his friend, a member of royalty, to become a consistent winner. He also developed Pascal's triangle, which you may remember from your early mathematics education. The statistical development of the normal or bell-shaped curve (see Chapter 4) is interesting. For many years, this development was attributed to Karl Friedrich Gauss (early 1800s), and was actually known for some time as the Gaussian curve. Later historians found that Abraham DeMoivre actually developed the normal curve in the 1730s. As statistics was not thought of as a true academic discipline until the late 1800s, people like Pascal and DeMoivre were consulted by the wealthy on odds about games of chance and by insurance underwriters to determine mortality rates.

Karl Pearson is one of the most famous statisticians to date (late 1800s to early 1900s). Among his many accomplishments is the Pearson product-moment correlation coefficient still in use today (see Chapter 10). You may know of Florence Nightingale (1820–1910) as an important figure in the field of nursing. However, you may not know of her importance in the field of statistics. Nightingale believed that statistics and theology were linked and that by studying statistics we might come to understand God's laws.

A quite interesting statistical personality is William Sealy Gossett, who was employed by the Guinness Brewery in Ireland. The brewery wanted to select a sample of people from Dublin in 1906 for purposes of taste testing. Gossett was asked how large a sample was needed in order to make an accurate inference about the entire population (see next section). The brewery would not let Gossett publish any of his findings under his own name,

so he used the pseudonym of Student. Today the *t* distribution is still known as Student's *t* distribution. Sir Ronald A. Fisher is another of the most famous statisticians of all time. Working in the early 1900s Fisher introduced the analysis of variance (see Chapters 11–16) and Fisher's *z* transformation for correlations (see Chapter 10). In fact, the major statistic in the analysis of variance is referred to as the *F* ratio in honor of Fisher. These individuals represent only a fraction of the many famous and interesting statisticians over the years. For further information about these and other statisticians, we suggest you consult the references noted previously (e.g., Heyde et al., 2001; Pearson, 1978; Stigler, 1986), which provide many interesting stories about statisticians.

1.3 General Statistical Definitions

In this section we define some of the most basic concepts in statistics. Included here are definitions and examples of the following concepts: population, parameter, sample, statistic, descriptive statistics, and inferential statistics.

The first four concepts are tied together, so we discuss them together. A **population** is defined as *all members of a well-defined group*. A population may be large in scope, such as when a population is defined as all of the employees of IBM worldwide. A population may be small in scope, such as when a population is defined as all of the IBM employees at the building on Main Street in Atlanta. *The key is that the population is well defined* such that one could determine specifically who all of the members of the group are and then information or data could be collected from all such members. Thus, if our population is defined as all members working in a particular office building, then our study would consist of collecting data from all employees in that building. It is also important to remember that *you*, the researcher, define the population.

A **parameter** is defined as a *characteristic of a population*. For instance, parameters of our office building example might be the number of individuals who work in that building (e.g., 154), the average salary of those individuals (e.g., $49,569), and the range of ages of those individuals (e.g., 21 to 68 years of age). When we think about characteristics of a population we are thinking about **population parameters**. The two terms are often linked together.

A **sample** is defined as a *subset of a population*. A sample may be large in scope, such as when a population is defined as all of the employees of IBM worldwide and 20% of those individuals are included in the sample. A sample may be small in scope, such as when a population is defined as all of the IBM employees at the building on Main Street in Atlanta and 10% of those individuals are included in the sample. Thus, a sample could be large or small in scope and consist of any portion of the population. *The key is that the sample consists of some, but not all, of the members of the population*; that is, anywhere from one individual to all but one individual from the population is included in the sample. Thus, if our population is defined as all members working in the IBM building on Main Street in Atlanta, then our study would consist of collecting data from a sample of some of the employees in that building. It follows that if we, the researchers, define the population, then we also determine what the sample will be.

A **statistic** is defined as a *characteristic of a sample*. For instance, statistics of our office building example might be the number of individuals who work in the building that we sampled (e.g., 77), the average salary of those individuals (e.g., $54,022), and the range of ages of those individuals (e.g., 25 to 62 years of age). Notice that the statistics of a sample

alphabet, the name of the symbol, and how the symbol is commonly used in statistics. The table also includes additional notation commonly used to denote statistics. We will use many of these symbols throughout the text. This table is provided with a caveat. Unfortunately, statistical notation is not standardized. Should you pick up a different text, it's likely that the authors have used at least some different notation than what has been used in this text. (Argh! How frustrating, right?) Thus, throughout the text we have attempted to clearly indicate what the notation means as it is used.

1.4 Types of Variables

There are several terms we need to define about variables. First, it might be useful to define the term variable. A **variable** is defined as *any characteristic of persons or things that is observed to take on different values*. In other words, the values for a particular characteristic vary across the individuals observed. For example, the annual salary of the families in your neighborhood varies because not every family earns the same annual salary. One family might earn $50,000 while the family right next door might earn $65,000. Thus, the annual family salary is a *variable* because it *varies* across families.

In contrast, a **constant** is defined as *any characteristic of persons or things that is observed to take on only a single value*. In other words, the values for a particular characteristic are the *same* for all individuals or units observed. For example, say every family in your neighborhood has a lawn. Although the nature of the lawns may vary, everyone has a lawn. Thus, whether a family has a lawn in your neighborhood is a constant and therefore would not be a very interesting characteristic to study. *When designing a study, you (i.e., the researcher) can determine what is a constant*. This is part of the process of *delimiting*, or narrowing the scope of, your study. As an example, you may be interested in studying career paths of girls who complete AP science courses. In designing your study, you are only interested in girls, and thus sex would be a constant—you would be delimiting your study to girls. This is not to say that the researcher wholly determines when a characteristic is a constant. It is sometimes the case that we find that a characteristic is a constant *after* we conduct the study. In other words, one of the measures has no variation—everyone or everything scored or remained the same on that particular characteristic.

A number of different typologies are available for describing variables. One typology is categorical (or qualitative) versus numerical (or quantitative), and within numerical, discrete and continuous. A **categorical variable** is a *qualitative variable that describes categories of a characteristic or attribute*. Examples of categorical variables include political party affiliation (Republican = 1, Democrat = 2, Independent = 3), religious affiliation (e.g., Methodist = 1, Baptist = 2, Roman Catholic = 3, etc.), and course letter grade (A = 4, B = 3, C = 2, D = 1, F = 0). A **dichotomous variable** (also known as a *binary variable*) is a special, restricted type of categorical variable and is defined as a *variable that can take on only one of two values*. For example, sex at birth is a variable that can take on the values of male or female and is often coded numerically as 0 (e.g., for males) or 1 (e.g., for females). Other dichotomous variables include pass/fail, true/false, living/dead, and smoker/non-smoker. Dichotomous variables will take on special importance if you later study binary logistic regression.

A **numerical variable** is a quantitative variable. Numerical variables can further be classified as either discrete or continuous. A **discrete variable** is defined as a *variable that can only take on certain values*. For example, the number of children in a family can only take on certain values. Many values are not possible, such as negative values (e.g., the Joneses cannot have −2 children) or decimal values (e.g., the Smiths cannot have 2.2 children). In contrast, a **continuous variable** is defined as a *variable that can take on any value within a certain range, given a precise enough measurement instrument*. For example, the distance between two cities can be measured in miles, with miles estimated in whole numbers. However, given a more precise instrument with which to measure, distance can even be measured down to the inch or millimeter. When considering the difference between a discrete and continuous variable, keep in mind that *discrete variables arise from the counting process* and *continuous variables arise from the measuring process*. For example, the number of students enrolled in your statistics class is a discrete variable. If we were to measure (i.e., count) the number of students in the class, it would not matter if we counted first names alphabetically from A to Z or if we counted beginning with the person sitting in the front row to the last person sitting in the back row—either way, we would arrive at the same value. In other words, how we "measure" (again, in this instance, how we count) the students in the class does not matter—we will always arrive at the same result. In comparison, the value of a continuous variable is dependent on how precise the measuring instrument is. Weighing yourself on a scale that rounds to whole numbers will give us one measure of weight. However weighing on another, more precise, scale that rounds to three decimal places will provide a more precise measure of weight.

Here are a few additional examples. Other discrete variables include number of books owned, number of credit hours enrolled, and number of teachers employed at a school. Other continuous variables include salary (from zero to billions in dollars and cents), age (from zero up, in millisecond increments), height (from zero up, in increments of fractions of millimeters), weight (from zero up, in increments of fractions of ounces), and time (from zero up, in millisecond increments). Variable type is a very important concept in terms of selecting an appropriate statistic, as will be shown later.

1.5 Scales of Measurement

Another concept useful for selecting an appropriate statistic is the scale of measurement of the variables. First, however, we define **measurement** as the *assignment of numerical values to persons or things according to explicit rules*. For example, how do we measure a person's weight? Well, there are rules that individuals commonly follow. Currently weight is measured on some sort of balance or scale in pounds or grams. In the old days weight was measured by different rules, such as the number of stones or gold coins. These explicit rules were developed so that there was a standardized and generally agreed upon method of measuring weight. Thus, if you weighted 10 stones in Coventry, England, then that meant the same as 10 stones in Liverpool, England.

In 1951 the psychologist S.S. Stevens developed four types of measurement scales that could be used for assigning these numerical values. In other words, the type of rule used was related to the measurement scale. The four types of measurement scales are the nominal, ordinal, interval, and ratio scales. They are presented in order of increasing complexity (i.e., *nominal is the simplest* and *ratio is the most complex*) and of increasing information

1.6 Additional Resources

A number of excellent resources are available for learning statistics. Throughout the text, we will introduce you to many related to the respective topics for the concepts studied in the individual chapters. Here we offer recommendations for resources that are a bit more general in nature for learning, understanding, and appreciating statistics:

- Designed as a reference tool for manuscript and proposal reviewers, this is a great tool for researchers learning about statistics and wanting to learn more about quantitative data analysis (Hancock & Mueller, 2010)
- A resource that introduces readers to statistical concepts through verse, graphics, and text, with no equations (Keller, 2006)
- An edited text whose contributions from authors address statistical issues (e.g., mediation), methodological issues (e.g., qualitative research, sample size practices), and more (Lance & Vandenberg, 2009)
- Statistical misconceptions related to, among others, probability, estimation, hypothesis testing, ANOVA, and regression are discussed and discarded (Huck, 2016)
- A great additional resource that explains statistics in plain language (Huck, 2012)
- Common statistical conventions, ranging from sample size to bootstrapping to transformations and just about everything in between (Van Belle, 2002)
- A dictionary of statistics and related terms (Vogt, 2005)

Problems

Conceptual Problems

1. A mental health counselor is conducting a research study on satisfaction that married couples have with their marriage. In this scenario, "Marital status" (e.g., single, married, divorced, widowed) is which of the following?

   a. Constant
   b. Variable

2. Belle randomly samples 100 library patrons and gathers data on the genre of the "first book" that they checked out from the library. She finds that 85 library patrons checked out a fiction book and 15 library patrons checked out a nonfiction book. Which of the following best characterizes the type of "first book" checked out in this study?

   a. Constant
   b. Variable

3. For interval-level variables, which of the following properties does *not* apply?

   a. A is two units greater than B.
   b. A is greater than B.
   c. A is twice as good as B.
   d. A differs from B.